EXACTNESS OF A COMPLEX OF QUASI-COHERENT MODULES DETECTED BY PARASITIC COHOMOLOGY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Comparison of module categories on the small etale site. Cohomology of Algebraic Stacks, Proposition stacks-cohomology-proposition-loc-qcoh-flat-base-change. Identifies Ker/Im computed in the two topologies.
- Flat pullback of quasi-coherent modules is exact. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-flat-pullback-quasi-coherent; Sheaves on Stacks, Lemma stacks-sheaves-lemma-compare-quasi-coherent. Exactness of the qc restriction composite.
- Checking a quasi-coherent module is zero. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-quasi-coherent-check-exact. Concludes $\mathrm{E} = 0$ from vanishing on flat points.

Source reference. Original section 0GQG: sources/stacks-chow/tags.txt, source lines 20861–20861. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 0771: sources/stacks-chow/tags.txt, source lines 9186–9186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 076X: sources/stacks-chow/tags.txt, source lines 9182–9182. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GQD: sources/stacks-chow/tags.txt, source lines 20858–20858. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GQF: sources/stacks-chow/tags.txt, source lines 20860–20860. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let $\alpha : \mathcal{F} \to \mathcal{G}$ and $\beta : \mathcal{G} \to \mathcal{H}$ be maps in $QCoh(\mathcal{O}_\mathcal{X})$ with $\beta \circ \alpha = 0$. The following are equivalent:*

1. *in the abelian category $QCoh(\mathcal{O}_\mathcal{X})$ the complex $\mathcal{F} \to \mathcal{G} \to \mathcal{H}$ is exact at $\mathcal{G}$,*
2. $\mathrm{Ker}(\beta)/\mathrm{Im}(\alpha)$ *computed in either $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ or $Mod(\mathcal{X}_{fppf}, \mathcal{O}_\mathcal{X})$ is parasitic.*

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Proof. We have $QCoh(\mathcal{O}_\mathcal{X}) \subset LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$, see Section 0GQG. Hence $\mathrm{Ker}(\beta)/\mathrm{Im}(\alpha)$ computed in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ or $Mod(\mathcal{X}_{fppf}, \mathcal{O}_\mathcal{X})$ agree, see Proposition 0771. From now on we will use the étale topology on $\mathcal{X}$.

Let $\mathcal{E}$ be the cohomology of $\mathcal{F} \to \mathcal{G} \to \mathcal{H}$ computed in the abelian category $QCoh(\mathcal{O}_\mathcal{X})$. Let $x : U \to \mathcal{X}$ be a flat morphism where U is a scheme. As we are using the étale topology, the restriction functor $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X}) \to Mod(U_{étale}, \mathcal{O}_U)$ is exact. On the other hand, by Lemma 076X and Sheaves on Stacks, Lemma 0GQD the restriction functor

$$QCoh(\mathcal{O}_\mathcal{X}) \xrightarrow{x^*} QCoh((Sch/U)_{étale}, \mathcal{O}) \xrightarrow{-|_{U_{étale}}} QCoh(U_{étale}, \mathcal{O}_U)$$

is exact too. We conclude that $\mathcal{E}|_{U_{étale}} = (\mathrm{Ker}(\beta)/\mathrm{Im}(\alpha))|_{U_{étale}}$.

If (1) holds, then $\mathcal{E} = 0$ hence $\mathrm{Ker}(\beta)/\mathrm{Im}(\alpha)$ restricts to zero on $U_{étale}$ for all U flat over $\mathcal{X}$ and this is the definition of a parasitic module. If (2) holds, then $\mathrm{Ker}(\beta)/\mathrm{Im}(\alpha)$ restricts to zero on $U_{étale}$ for all U flat over $\mathcal{X}$ hence $\mathcal{E}$ restricts to zero on $U_{étale}$ for all U flat over $\mathcal{X}$. This certainly implies that the quasi-coherent module $\mathcal{E}$ is zero, for example apply Lemma 0GQF to the map $0 \to \mathcal{E}$. □